2019 4th International Conference on Information Systems and Computer Networks (ISCON)
GLA University, Mathura, UP, India. Nov 21-22, 2019

Motorcycle Safety Solution using the Internet of Things

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***Abstract*—This research has been based on a review of the safe transportation facilities for motorcycle safety and other similar issues with the help of existing or emerging IT technology to enhance the person's safety. There are multiple technologies to provide safety, security & ease of access to cars, commercial vehicle, and public transportation services. Current technologies have been developed with private and commercial vehicles like cars and trucks because of the ease of access and bigger market shares. But there is no advancement in the safety, security, and ease of use of a motorcycle, even though the motorcycle market is great and big in some Asian countries, only a few it-enabled services and technologies have been developed for the safety of a motorcycle rider and those which have been developed are in-vehicle systems. Many it enabled technologies and services that have been or are being developed for other classes of vehicles, still, there is a requirement for the development of IT technologies for motorcycles, similar to the need for innovation for other modes of transport, motorcycle has a problem when it comes to implementation of any system, especially if it has not been designed for a motorcycle. This system provides rider safety along with multiple features which are not available on any other helmet available in the market using the benefits provided by the internet of things which helps in additional features other than the main purpose of a motorcycle helmet that is safety. The proposed model revolves around the rider's safety whilst providing ease of access and comfort. The helmet consists of a power module, a BLE for nearby detection, a geographical positioning system, a display system that does not interfere with the view of the wearer and an audio interface for audio communication and voice commands.**

Keywords— motorcycle, road safety, smart helmet, internet of things

I. INTRODUCTION

The easiest mode of transport has been proven to be a motorcycle, with its small size and high speed, it is preferred by many people as their daily driver although it is also the riskiest one. Motorcycles have proven to be a popular means of transportation, especially in Asian countries. In fact, 50% of Malaysian road users are registered as motorcycle owners. But because motorcycle riders are much more vulnerable road users, many road accidents involve them as victims. Many people tend to forget the most basic safety needed, which can prove to be helpful in many situations i.e. A helmet. As per govt. Reports [1], 1,57,723 two-wheeler accidents happened in India in 2017 which is nearly 35% of the total road deaths in 2017 (in India). Crashes due to two-wheelers resulted in death of 48,746 riders in road accidents, out of which 73.8% riders were not wearing a helmet. The main reasons for the vulnerability and high risk to the rider are due to the exposed body and very little or sometimes, no protection offered by motorcycle safety devices during an accident. And so, to reduce the severity of motorcyclist injuries and increasing the motorcycle safety there is a need to develop a technology which can ensure the safety of the rider. The smart helmet is designed to not let the motorcycle start unless the helmet registered is worn and is nearby (within a meter) and also to turn the engine off in case the helmet is taken off after ignition. Apparently, many riders think of a helmet as a useless and boring thing until they actually need one. The helmet is designed for comfort, safety, accessibility, and integration with the motorcycle via internet communication enabled.

II. LITERATURE REVIEW

Ministry of road transport & highway safety states that road transport is essential for improvement as it offers mobility to people and items. But it also exposes them to the chance of collisions. The traffic surroundings of India are getting worse due to the high number of motor vehicles fueled through an excessive fee of monetary increase. Today, road injuries are one of the main causes of death, disabilities and hospitalization in India. Giannopoulos, d. Vlachos states that finite detail evaluation of effect damage response of composite motorcycle protection helmets robin Gobelin, Donald Varner James j. Snyder, James parsons' states that the existing invention relates to stopping systems for car use and, greater specially, to an improved helmet-set up auxiliary stop system especially perfect for use with a bike or ridden car. Jorge Batista presents a framework that mixes a strong facial feature place with an elliptical face modeling to degree a driver's vigilance stage. The proposed answer offers with the computation of eyelid motion parameters and head (face) factors of attention. M. Bayley, m. Regan & s. Hosking states that the present day observe aimed to discover present and rising wise transport systems (its) that have the capability to beautify bike rider safety. An evaluation of the literature discovered that only a few commercially available it's presently exists in particular for

motorcycles, although numerous emerging technologies were recognized. Consultations with global experts in its, motorbike protection, motorbike manufacturers, and diverse road protection research groups showed this

III. THE PROBLEM

The traditional helmet is designed to save the head of the motorcycle rider in an event of a collision but may riders tend to leave it behind which is has a big hand in the number of permanent damages. Other problems discussed in the paper are stated below.

A. *Safety*

The research has been conducted in light of the problem of safety of one who rides a motorcycle and does not wear a helmet. People tend to forget wearing helmets or leave the helmets behind due to peer pressure and to look "cool" but they do not think once about the benefits they would have in case of an accident.

B. *Exceptions and non-lethal emergencies*

Not wearing the helmet, the motorcycle module will not allow the ignition to start without the presence of the helmet. It will, however, have a panic walk around in case of emergencies whilst notifying the registered users and their secondary users (i.e. Partner, family, etc.) By sending the GPs location to the servers for added safety.

C. *View while riding*

The motorcycle rider's attention is distracted by other factors like calls, uncommon sounds from the motorcycle, no directional idea and navigational devices being on uneasy locations in the field of view of the motorcycle rider.

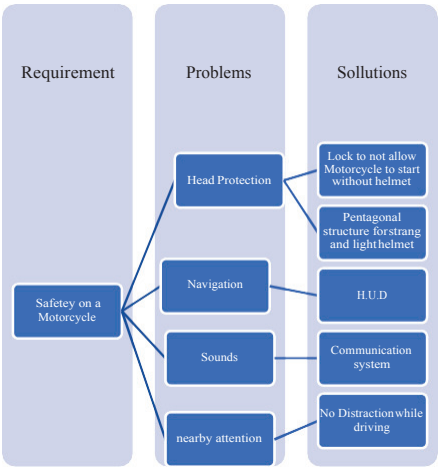

Fig.1. The research paper outline

D. *Audible distractions*

While riding a motorcycle if there are any incoming calls on their mobile phones, the motorcyclists have to stop the vehicle and pick up the phone in order to communicate, but this process requires time and can be destructive in some situations, some motorcyclists prefer to wear earphones in order to facilitate this process which can be dangerous for the environmental alertness of the rider as earphones can suppress sounds from other vehicles nearby which may lead to life-threatening accidents

IV. THE PROPOSED SOLUTION

The proposed model has a processing module that is required to ensure the rigidity fit and distance from the motorcycle itself for activation of the ignition when in range and to turn off the engine in event of distancing itself from the motorcycle module

A. *Capacitive sensors*

The capacitive sensors fixed in the helmet can detect whether there is proper contact with the skin of the person wearing the helmet and detecting whether the helmet is worn the way it requires to be worn.

Gainsaying the problem, the helmet has three capacitance sensing modules that help in detecting if the helmet is worn effectively or not ensuring the rider's safety and tackling the initial problem.

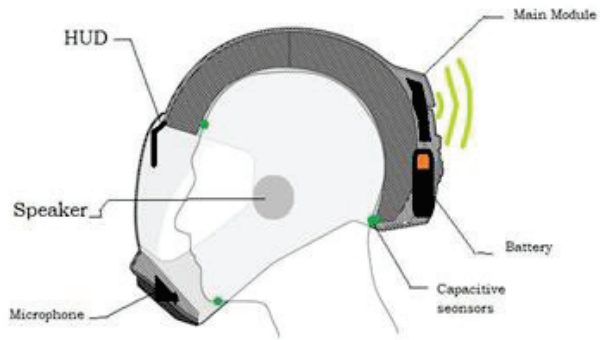

Fig. 2. The helmet with components

B. *Heads Up Display (H.U. D)*

The HUD will enable the user to navigate, use other applications like a clock, reminder, nearby search, assistant, etc. The HUD will also display the statistics of the motorcycle itself, the battery on the helmet, and the battery left on the rider's mobile phone. All other information will be displayed to the rider through the same HUD. While being over the rider's field of view the HUD is placed in a position where it does not obstruct the view at all, the HUD is on the side of the field of view and shows a translucent image of the content to be shown with least possible distractions and good navigation and safety. The HUD module is placed in a way that even in case of a head-on collision the HUD itself cannot shatter and cannot harm the rider, following the principles and methodology of Intel's vaunt [2] project, the HUD can display over the view of the rider without distracting the rider

C. *Audio communication devices*

The speaker and microphone modules work together ensuring the least noise reduction and maximum ease of use while traveling, the rider can play music, talk to someone over

the phone, listen to navigation and it will not obstruct the nearby necessary noise from the traffic. Using low voltage drivers to ensure the least coverage over the nearby noise, a communication system can be built to connect two or more devices in a conference connection via the Internet.

V. THE MOTORCYCLE MODULE

With the help of a BLE module, the motorcycle module can detect the device in the helmet if it is within a short range less than 2 meters. The motorcycle module has to be fitted inside the motorcycle in a secure place under the rider's seat for the best work environment. The module consists of an antenna, a processing unit with BLE enabled connectivity and a subscriber identifying module an operating module and access to the ignition of the whole motorcycle through its circuit. The module is designed to integrate with the motorcycle whilst keeping the motorcycle connected to the Internet for details of the conditions and risks in the operation of the motorcycle. The module sends data to the rider when there is a need for maintenance and possible future problems.

The signal from the motorcycle module keeps scanning nearby compatible devices, when a registered helmet approaches the signal area, the motorcycle stops scanning for devices and starts scanning for a "green signal" from the helmet.

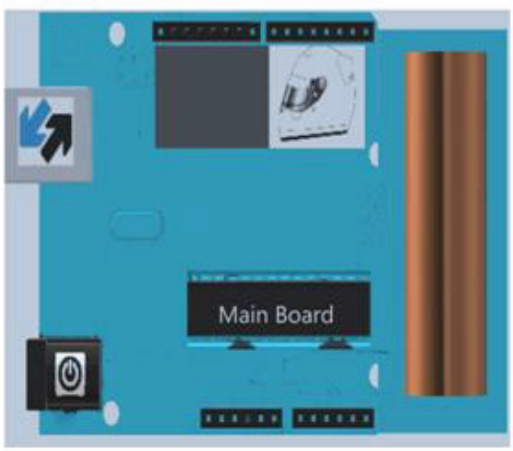

Fig. 3 The motorcycle module

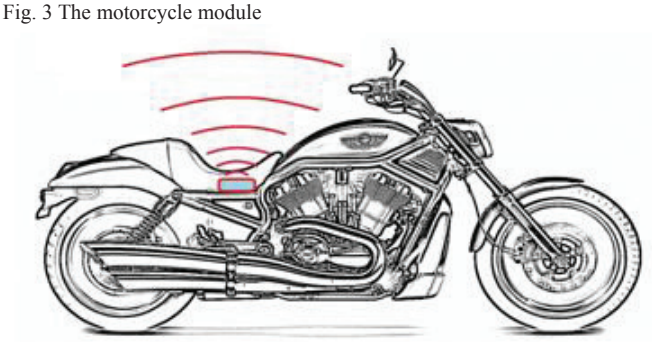

Fig. 4. Motorcycle module positioning.

Once the green signal is received, the motorcycle module enables ignition and starts scanning for a red signal. If the rider removes the helmet at this stage, the helmet waits for a time of 5 seconds and sends a red signal to the motorcycle module unless the capacitive sensors fixed in the helmet receive a positive response within the stipulated time frame. Once the motorcycle module receives a red signal, it gives a time of 20 seconds to the rider to slow down and drive in a safe place before the motorcycle module turns the engine off and disables the ignition.

The Module is also responsible for the data processing and transmission which will be collected form the motorcycle sensors. The module then will send the information in a specific format to the server via internet connection enabled using SIM cards. The module will send a message to the server with the details of the motorcycle including Fuel level, Oil Levels, engine temperature, maintenance requirements etc. The module also sends the data to the Helmet module using the Bluetooth connection available for constant monitoring.

The information is then displayed to the user wearing the helmet via the H.U.D inside the helmet for safety and monitoring of the probable causes of errors. The helmet module also ensures the rigid fit of the helmet on the head of the wearer and enables collision alarm and disables the motorcycle ignition and engine rev.

VI. THE STRUCTURAL COMPOSITION OF THE HELMET

The most basic requirements of the project are the helmet hardware itself, the whole helmet is designed to ensure safety in the worst case, for which the whole helmet is required to be is certified. The impact force absorption in the midst of impact to the rider is reliable of fundamental noteworthiness in order to guarantee the rider against the head home in the midst of an accident. In the present survey, a parametric investigation has been performed in order to examine the effect of the composite shell tightness and damage avoidance in the midst of impact, on the dynamic response of the composite rider safety helmet. This kind of parametric examination may be used as a key aspect in the midst of designing a helmet.

The express hydrodynamic method was used to explore an ordered model of the helmet head structure system to re-enact its dynamic response in the event of an impact. A gigantic bit of the work was revolved around of the mechanical lead of the composite materials, including damage and delamination improvement. The dynamic response of the different defensive top head structure, systems was settled on a choice to the extent the best reviving saw at the point of convergence of force of impact and the most extraordinary estimation of head injury expected. It was seemed composite shell structures demonstrating lower shear execution give additional imperativeness holding frameworks and result in better crashworthiness helmet behavior.

VII. CONCLUSION

Statistics display that motorcycle rider is more at risk than any other type of road user. In addition, although the number of road accidents has decreased in recent years, the number of motorcycle fatalities has increased. The motorcycle market is huge, so is the number of motorcycle riders and the number of motorcycle accidents. Some situations might be avoided, but it is always better to be prepared for any mishap while driving or riding a motorcycle on the road.

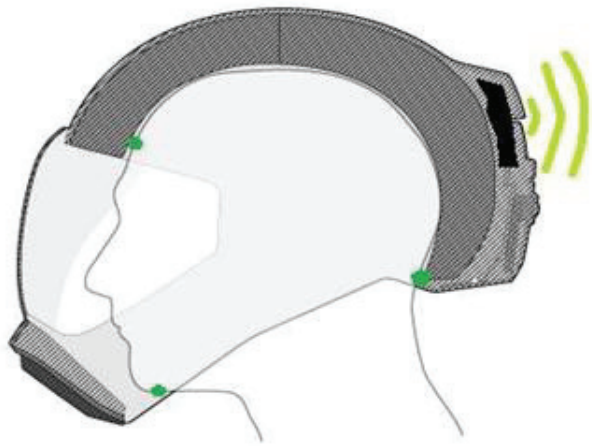

Fig. 5. The helmet

A helmet is a necessity while riding at a dangerous speed, exposing the whole body of the person riding the motorcycle other safety gears are also advisable but cannot be worn all the time because of the heavy weight and bulky sizes of the safety gear, a helmet can be taken anywhere and is not baggage to take care of even while not riding, and because the hands of a motorcycle rider are busy while riding, we cannot assign more controls in the hands of the person, we need to assign the controls to through voice commands. And to enable that we need a system within the helmet as there is too much wind and noise when one rides a motorcycle. The only safety for the head can also be designed to enable other features like navigation, voice communication, entertainment system for the long run, etc. There are multiple possible features that can be embedded if there is the required hardware. The solution if implemented enables seamless connectivity between the motorcycle and the motorcycle rider. The motorcycle module can enable the rider to have real time information about the temperature of the engine, oil levels, fuel levels, tire pressure, distance till the next required stop, navigation, all without being distracted from the riding process and without having to take eyes off the road the helmet, it needs to be strong enough to save the rider from the force of impact in an event of an accident and being able to do so while housing the whole system in a way that is does not interfere with the comfort of the rider because an uncomfortable helmet is not preferred by most of the riders due to the distractions it can create.

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